

DS4 Talker

LAB 4

SECTION H

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Questions

1. Each word was read using `fgets()` after the file was opened with `fopen()`, they were stored in a buffer, and then cleaned it using a trim function, `trimw()`, to remove any extra spaces or newlines at the end. After it was trimmed, it was then copied into an array so each word had its own location, using `strcpy()`. I repeated this until all words were read or I reached the maximum limit, if the end of the file was reached `fclose()` was used.
2. Keeping track of the selected word on screen was done by using an integer variable, `selectedIndex`, which stored the index of the currently selected word in the array. The words were displayed in 5 columns, the position of the screen was based on this index. When words were printed with `mvprintw()`, it would also print a marker, to the word whose index matched with `selectedIndex`. When the joystick moved, the program changed `selectedIndex` by subtracting or adding. To select a new word in a different row and or column, while also making sure to keep within the bounds. Also, every time `selectedIndex` changes, the code redraws the cursor into the new position, so that it is always clear which word is currently being selected.
3. Knowing how long each word added to the sentence, I can use this info to delete the last thing added. I would have an array, where every time something is appended to the sentence it stores how many characters it added. I also, had a sentence buffer, as well as an integer to store the current length of the sentence. When I wanted to delete the last item, the program would look at the last value in the array, and subtract that length from the sentence ending things off with a `'\0'` to mark the end of the string. It also decreased the counter, and redrew the sentence.
4. After experiencing this lab, I would personally say that the joystick was easiest to use for movement. In my head it just makes more sense, and also, each joystick give values for x and y that easily display it's directions, left, right, up, and down. To implement them, you can use a threshold, and see if the values go above it or not to determine weather or not to move the cursor. With the gyro, there is so much interpret with the data, and the sensitivity can be vastly different amongst controllers making it difficult to test. The values the gyro gives aren't very straight forward, which can cause some confusion while trying to programming to do a specific function. I also feel a part of it, is growing up using consoles, most games were played using the joysticks and not the gyro, so there was some level of comfortability with it.